

HW2

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```
set.seed(123)
```

Question 1

Task 3

Write a function that takes as its inputs that data you simulated (or any data of the same type) and a sequence of θ values of length 1000 and produces likelihood values based on the Binomial Likelihood. Plot your sequence and its corresponding Likelihood function.

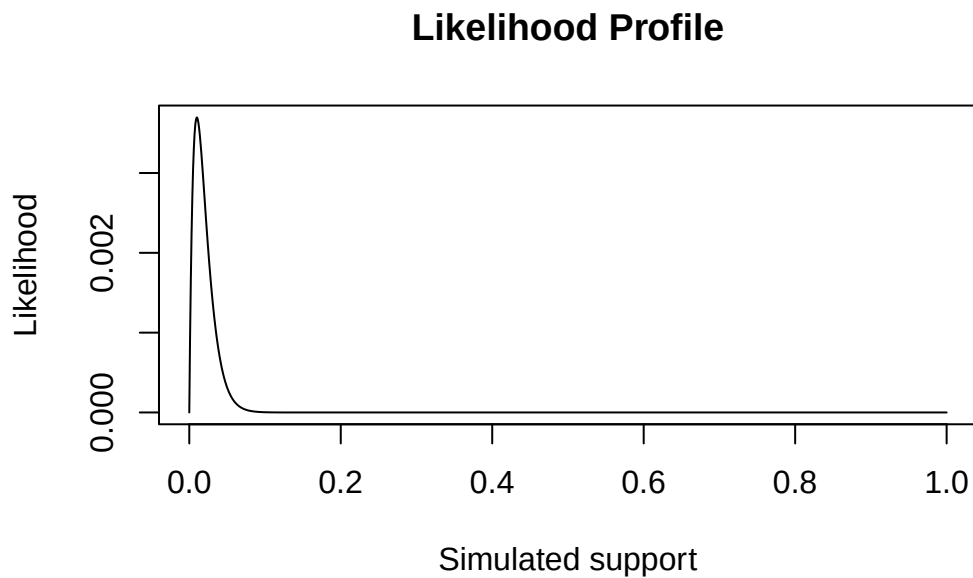
The likelihood function is given below. Since this is a probability and is only valid over the interval from $[0, 1]$ we generate a sequence over that interval of length 1000.

You have a rough sketch of what you should do for this part of the assignment. Try this out in lab on your own.

```
# define observed data
obs_data <- rbinom(n=100, size=1, prob=0.01)
### Bernoulli LH Function ###
# Input: obs_data, theta
# Output: bernoulli likelihood
bernoulli_lh <- function(obs_data, theta) {
  n<-length(obs_data) # define num samples
  x<-sum(obs_data) # define sample data sum
  return(theta^x*(1-theta)^(n-x))
}

### Plot LH for a grid of theta values ###
# Create the grid #
thetas = seq(0, 1, length.out = 1000)
```

```
# Store the LH values
lh_values<- bernoulli_lh(obs_data, thetas)
# Create the Plot
plot(thetas, lh_values, type = "l", main= "Likelihood Profile",
      xlab="Simulated support",ylab = "Likelihood")
```



Task 4

Write a function that takes as its inputs prior parameters a and b for the Beta-Bernoulli model and the observed data, and produces the posterior parameters you need for the model. **Generate and print** the posterior parameters for a non-informative prior such as $(a,b) = (1,1)$ and for an informative case $(a,b) = (3,1)$.

```
# function to update prior params for Beta-Bernoulli
## inputs: observed data, a, b
update_params <- function(obs_data, a, b) {
  n <- length(obs_data) # define num samples
  x <- sum(obs_data) # define sample data sum

  results <- list(a_param = x + a, b_param = n - x + b) # append updated params
  return(results)
```

```

}
# define non-informative a,b
a1<-1
b1<-1
# define informative a,b
a2<-3
b2<-1

# run function for non-info case
output1 <- update_params(obs_data, a1, b1)
# extract a,b
a1_new<- output1$a_param
b1_new<- output1$b_param
# print
print(a1_new)

```

```
[1] 2
```

```
print(b1_new)
```

```
[1] 100
```

```

# run function for info case
output2 <- update_params(obs_data, a2, b2)
# extract a,b
a2_new<- output2$a_param
b2_new<- output2$b_param
# print
print(a2_new)

```

```
[1] 4
```

```
print(b2_new)
```

```
[1] 100
```

Here we wrote a function *update_params* that takes as its inputs prior parameters a and b for the Beta-Bernoulli model and the observed data, *obs_data*, and produces the posterior parameters we need for the model. The posterior parameters for a non-informative prior $(a, b) = (1, 1)$ were $(a_n, b_n) = (2, 100)$, and the posterior parameters for an informative case $(a, b) = (3, 1)$ were $(a_n, b_n) = (4, 100)$.

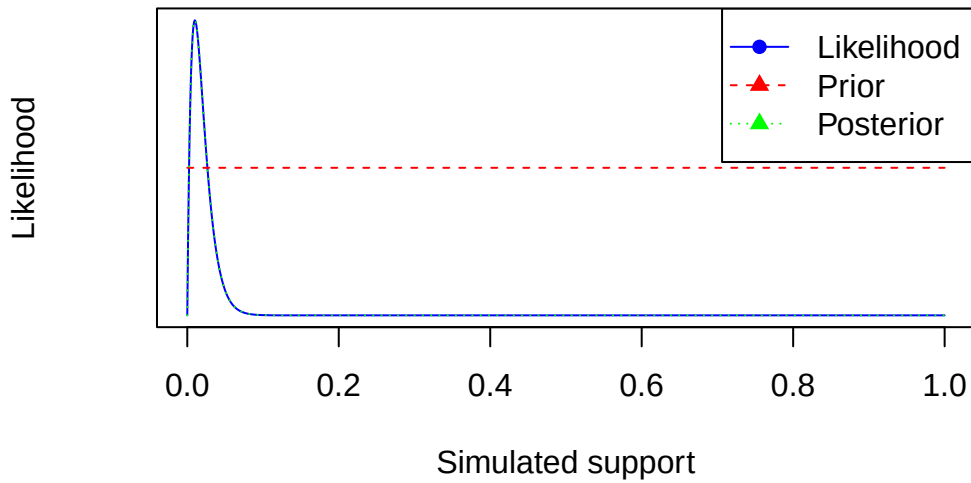
Task 5

Create two plots, one for the informative and one for the non-informative case to show the posterior distribution and superimpose the prior distributions on each along with the likelihood. What do you see? Remember to turn the y-axis off using the command `yaxt = "none"` in the `plot()` command. Why? Superimposing the distributions may make the scale non-sense.

```
# Non-informative case
## define prior
prior1 <- dbeta(thetas, shape1 = a1, shape2 = b1)
## define posterior using updated params
posterior1 <- dbeta(thetas, shape1 = a1_new, shape2 = b1_new)

## plot the likelihood
plot(thetas, lh_values, type = "l", col = "blue", pch = 16,
      xlab = "Simulated support", ylab = "Likelihood",
      main = "Non-informative prior", yaxt = "n",
      xlim = range(thetas), yaxt="none")
# overlay prior
par(new = TRUE)
# plot prior
plot(thetas, prior1, type = "l", col = "red", pch = 17, lty = 2,
      axes = FALSE, xlab = "", ylab = "", xlim = range(thetas),
      yaxt="none")
# overlay posterior
par(new = TRUE)
# plot posterior
plot(thetas, posterior1, type = "l", col = "green", pch = 17, lty = 3,
      axes = FALSE, xlab = "", ylab = "", xlim = range(thetas), yaxt="none")
# add legend
legend("topright", legend = c("Likelihood", "Prior", "Posterior"),
      col = c("blue", "red", "green"), lty = c(1, 2, 3), pch = c(16, 17, 17))
```

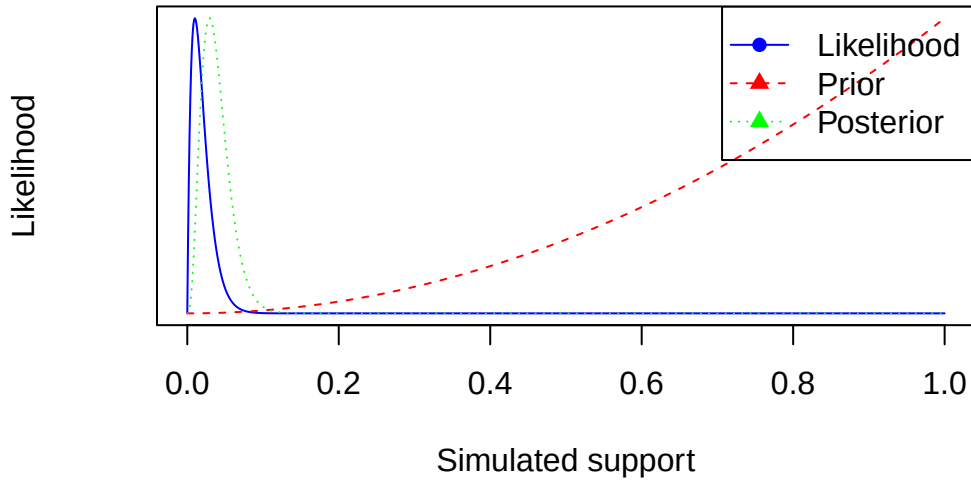
Non-informative prior



```
# Informative Case
# define prior
prior2 <- dbeta(thetas, shape1 = a2, shape2 = b2)
# define posterior with updated parameters
posterior2 <- dbeta(thetas, shape1 = a2_new, shape2 = b2_new)

# plot the likelihood
plot(thetas, lh_values, type = "l", col = "blue", pch = 16,
      xlab = "Simulated support", ylab = "Likelihood", main = "Informative prior",
      yaxt = "n", xlim = range(thetas), yaxt = "none")
# overlay prior
par(new = TRUE)
# plot prior
plot(thetas, prior2, type = "l", col = "red", pch = 17, lty = 2,
      axes = FALSE, xlab = "", ylab = "", xlim = range(thetas), yaxt = "none")
# overlay posterior
par(new = TRUE)
# plot posterior
plot(thetas, posterior2, type = "l", col = "green", pch = 17, lty = 3,
      axes = FALSE, xlab = "", ylab = "", xlim = range(thetas), yaxt = "none")
# add legend
legend("topright", legend = c("Likelihood", "Prior", "Posterior"),
      col = c("blue", "red", "green"), lty = c(1, 2, 3), pch = c(16, 17, 17))
```

Informative prior



```
### Here, I used ChatGPT to help with plot formatting/styles
### comments are my own
```

Here, we see the posterior overlap the likelihood in the case of the non-informative prior. When we implement an informative prior, the posterior moves away from the likelihood and approaches the prior.

Question 2

Part A

Derive (step by step) the posterior density, $p(\theta|x_{1:n})$. Give the form of the posterior in terms of one of the most common distributions (Bernoulli, Beta, Exponential, or Gamma).

Proof. Suppose we have data $x_1, \dots, x_n$ which we are modeling as i.i.d. observations from an Exponential distribution with unknown parameter θ . We write the likelihood as

$$p(x_i | \theta) = \theta \exp(-\theta x_i), \quad x_i > 0.$$

Also, suppose the prior distribution is

$$\theta \sim \text{Gamma}(a, b),$$

where a is the shape parameter and a is the rate parameter. The prior density is

$$p(\theta) = \frac{b^a}{\Gamma(a)} \theta^{a-1} \exp(-b\theta), \quad \theta > 0.$$

We will find the posterior density $p(\theta \mid x_{1:n})$. By Bayes' theorem,

$$p(\theta \mid x_{1:n}) = \frac{p(\theta, x_{1:n})}{p(x_{1:n})} = \frac{p(x_{1:n} \mid \theta)p(\theta)}{p(x_{1:n})}.$$

Because $p(x_{1:n})$ does not depend on θ , it is absorbed into the constant of proportionality:

$$p(\theta \mid x_{1:n}) \propto p(x_{1:n} \mid \theta)p(\theta).$$

The likelihood is given by

$$\begin{aligned} p(x_{1:n} \mid \theta) &= \prod_{i=1}^n p(x_i \mid \theta) \\ &= \prod_{i=1}^n [\theta \exp(-\theta x_i)] \\ &= \theta^n \exp\left(-\theta \sum_{i=1}^n x_i\right). \end{aligned}$$

Combining the likelihood and prior gives

$$\begin{aligned} p(\theta \mid x_{1:n}) &\propto \left[\theta^n \exp\left(-\theta \sum_{i=1}^n x_i\right) \right] \left[\frac{b^a}{\Gamma(a)} \theta^{a-1} \exp(-b\theta) \right] \\ &\propto \theta^{a+n-1} \exp\left[-\theta \left(b + \sum_{i=1}^n x_i\right)\right]. \end{aligned}$$

This is the kernel of a Gamma distribution, with shape parameter $a + n$ and rate parameter $b + \sum_{i=1}^n x_i$. Therefore,

$$\theta \mid x_{1:n} \sim \text{Gamma}\left(a + n, b + \sum_{i=1}^n x_i\right).$$

Thus, the posterior density is

$$p(\theta \mid x_{1:n}) = \frac{(b + \sum_{i=1}^n x_i)^{a+n}}{\Gamma(a+n)} \theta^{a+n-1} \exp \left[-\theta \left(b + \sum_{i=1}^n x_i \right) \right], \quad \theta > 0.$$

QED

Part B

Why is the posterior distribution a proper density or probability distribution function?

The posterior distribution is a proper density/probability distribution function because it takes the form of an updated Gamma distribution, where $a' = a + n > 0$ and $b' = b + \sum_{i=1}^n x_i > 0$. The Gamma distribution is a known, common probability distribution. The posterior distribution is proper because multiplying the kernel (shown in **Part A**) by the normalizing constant ensures its density integrates to 1 over the entire parameter space, Θ .

Part C

Now, suppose you are measuring the number of seconds between lightning strikes during a storm, your prior is $\text{Gamma}(0.1, 1.0)$, and your data is

$$(x_1, \dots, x_8) = (20.9, 69.7, 3.6, 21.8, 21.4, 0.4, 6.7, 10.0).$$

Plot the prior and posterior p.d.f.s. (Be sure to make your plots on a scale that allows you to clearly see the important features.)

```
# define thetas (updated based on resulting scale)
thetas<-seq(0.001, 0.15, length.out = 1000)
# define data
data <- c(20.9, 69.7, 3.6, 21.8, 21.4, 0.4, 6.7, 10.0)
# define gamma distribution (prior)
a <- 0.1
b <-1.0
prior <- dgamma(x = thetas, shape = a, rate = b)
# define the posterior dist
n <- length(data) # define length of data
posterior <- dgamma(x = thetas, shape = a+n, rate = b + sum(data))

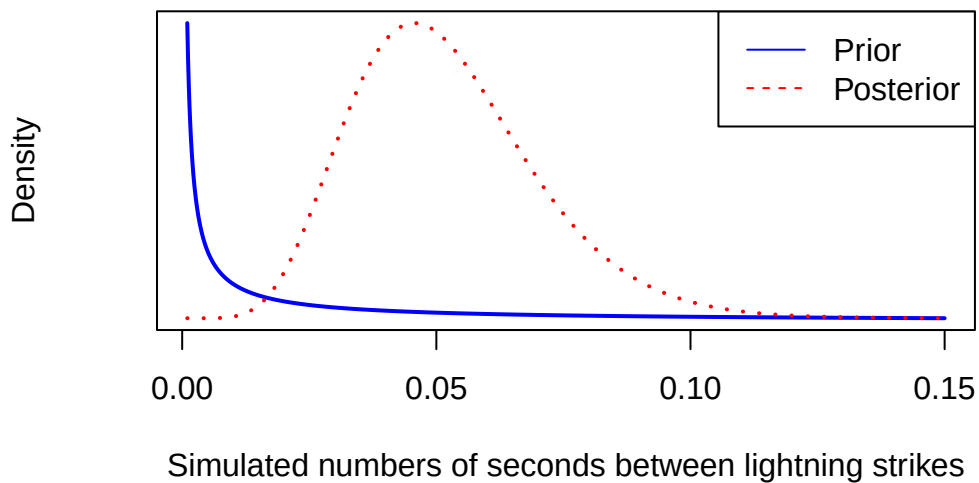
# plot prior
plot(thetas, prior,type = "l", col = "blue",lwd=2,
```



```

      xlab = "Simulated numbers of seconds between lightning strikes",ylab = "Density", yaxt =
# overlay posterior
par(new = TRUE)
# plot posterior
plot(thetas, posterior,type = "l", col = "red", lty = 3, lwd=2,
      axes = FALSE, xlab = "", ylab = "",xlim = range(thetas),yaxt="none")
# add legend
legend("topright",legend = c("Prior", "Posterior"),
      col = c("blue", "red"), lty = c(1, 2))

```



Part D

Give a specific example of an application where an Exponential model would be reasonable. Give an example where an Exponential model would NOT be appropriate, and explain why.

Reasonable: The Exponential model is often used to model the time between events. Thus, an appropriate application may be to model the wait time between customers served at a store, assuming a constant rate and independence of events, as well as the probability of the next event occurring not depending on how much time has already elapsed.

Unreasonable: One example where an exponential model would not be appropriate is for modeling coin flips. A coin flip has discrete, binary outcomes and exponential distributions model continuous outcomes.

Question 3

Priors, Posteriors, Predictive Distributions (Hoff, 3.9) An unknown quantity $Y|\theta$ has a Galenshore(a, θ) distribution if its density is given by

$$p(y|\theta) = \frac{2}{\Gamma(a)} \theta^{2a} y^{2a-1} e^{-\theta^2 y^2}$$

for $y > 0, \theta > 0, a > 0$. Assume for now that a is known and θ is unknown and a random variable. For this density,

$$E[Y] = \frac{\Gamma(a + 1/2)}{\theta \Gamma(a)}$$

and

$$E[Y^2] = \frac{a}{\theta^2}.$$

Part A

Identify a class of conjugate prior densities for θ . Assume the prior parameters are c and d , which are fixed and known. That is, state the prior distribution for θ , which will have known and fixed parameters c, d such that the resulting posterior is conjugate. Plot a few members of this class of densities.

Suppose θ also has Galenshore prior density, where $\theta \sim \text{Galenshore}(c, d)$. The prior parameters are c and d , which are fixed, known, and greater than 0. So, we have

$$p(\theta|c, d) = \frac{2}{\Gamma(c)} d^{2c} \theta^{2c-1} e^{-d^2 \theta^2} \quad \theta > 0$$

. We will show that the posterior density, $p(\theta|y_{1:n})$, is conjugate (i.e., the posterior takes on an updated form of the prior). Let $y_1, \dots, y_n$ be independent observations such that $Y_i|\theta \sim \text{Galenshore}(a, \theta)$. By Bayes' Theorem,

$$p(\theta|y_{1:n}) = \frac{p(\theta, y_{1:n})}{p(y_{1:n})} = \frac{p(\theta|y_{1:n})p(\theta)}{p(y_{1:n})}$$

. Since we are conditioning on the data, it is a constant, so we have

$$\frac{p(\theta|y_{1:n})p(\theta)}{p(y_{1:n})} \propto p(\theta|y_{1:n})p(\theta)$$

. Then, substituting in our likelihood and prior gives the following:

$$p(\theta|y_{1:n})p(\theta) = \prod_{i=1}^n \left(\frac{2}{\Gamma(a)} \theta^{2a} y_i^{2a-1} e^{-\theta^2 y_i^2} \right) * \left(\frac{2}{\Gamma(c)} d^{2c} \theta^{2c-1} e^{-d^2 \theta^2} \right)$$

. Combining terms and absorbing terms which do not depend on θ into the constant of proportionality yields

$$p(\theta | y_{1:n}) \propto \theta^{2an} \exp \left(-\theta^2 \sum_{i=1}^n y_i^2 \right) \theta^{2c-1} \exp(-d^2 \theta^2) = \theta^{2(c+na)-1} \exp \left[-\theta^2 \left(d^2 + \sum_{i=1}^n y_i^2 \right) \right].$$

. Therefore, we found $p(\theta|y_{1:n}) \propto \text{Galenshore}(\theta|c' = c + na, d' = \sqrt{d^2 + \sum_{i=1}^n y_i^2})$, so the resulting posterior is conjugate.

Here, we plot a few members of this class of densities:

```
# Galenshroe prior density function
# inputs: c,d, thetas
# output: priord density for each theta
galenshore_prior <- function(c, d, thetas) {
  # galenshore density
  2 / gamma(c) * d^(2 * c) * thetas^(2 * c - 1) * exp(-d^2 * thetas^2)
}
# create a grid of theta values
thetas <- seq(0.001, 3, length.out = 1000)

# deifne some different c,d values for comparisons of diff members of class
c1 <- 1
d1 <- 1

c2 <- 2
d2 <- 1

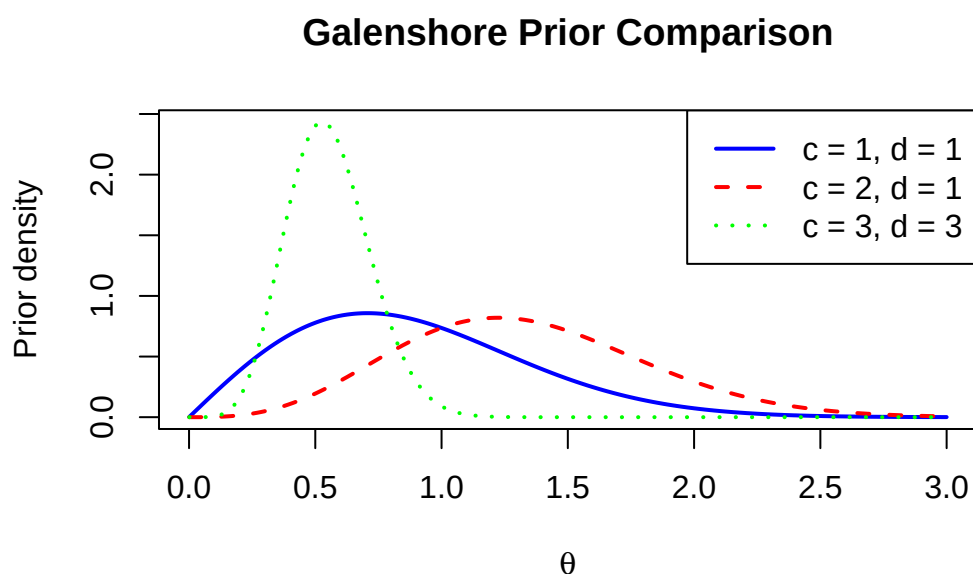
c3 <- 3
d3 <- 3

# compute prior densities usign function above
prior1 <- galenshore_prior(c1, d1, thetas)
prior2 <- galenshore_prior(c2, d2, thetas)
prior3 <- galenshore_prior(c3, d3, thetas)
```

```
# plot priors and overlay
plot(thetas, prior1, type = "l", col = "blue", lwd = 2, xlab = expression(theta),
     ylab = "Prior density", main = "Galenshore Prior Comparison",
     ylim = c(0, max(prior1, prior2, prior3)))

lines(thetas, prior2, col = "red", lwd = 2, lty = 2)
lines(thetas, prior3, col = "green", lwd = 2, lty = 3)

legend("topright", legend = c("c = 1, d = 1", "c = 2, d = 1", "c = 3, d = 3"),
     col = c("blue", "red", "green"), lty = c(1, 2, 3), lwd = 2)
```



Here, I used ChatGPT to help me fix formatting for long equations and to help me plot the densities.

Part B

Let $Y_1, \dots, Y_n \stackrel{iid}{\sim} \text{Galenshore}(a, \theta)$. Find the posterior distribution of $\theta | y_{1:n}$ using a prior from your conjugate class.

Let $\theta \sim \text{Galenshore}(c = 1, d = 1)$. By the proof-sketch shown in **Part A**, we have that

$\theta | y_{1:n} \sim \text{Galenshore}(1 + na, \sqrt{1^2 + \sum_{i=1}^n y_i^2})$, so

$$p(\theta \mid y_{1:n}) = \frac{2}{\Gamma(1+na)} \left(1 + \sum_{i=1}^n y_i^2\right)^{1+na} \theta^{2(1+na)-1} e^{-(1+\sum_{i=1}^n y_i^2)\theta^2} \quad \theta > 0$$

Part C

Show that

$$\frac{p(\theta_a \mid y_{1:n})}{p(\theta_b \mid y_{1:n})} = \left(\frac{\theta_a}{\theta_b}\right)^{2(an+c)-1} e^{(\theta_b^2 - \theta_a^2)(d^2 + \sum y_i^2)},$$

where

$$\theta_a, \theta_b \sim \text{Galenshore}(c, d).$$

Identify a sufficient statistic.

Let

$$\theta_a, \theta_b \sim \text{Galenshore}(c, d).$$

From **Part B**, we know

$$p(\theta_a \mid y_{1:n}) = \frac{2}{\Gamma(c+na)} (d^2 + \sum_{i=1}^n y_i^2)^{c+na} \theta_a^{2(c+na)-1} e^{-(d^2 + \sum_{i=1}^n y_i^2)\theta_a^2} \quad \theta_a > 0$$

and

$$p(\theta_b \mid y_{1:n}) = \frac{2}{\Gamma(c+na)} (d^2 + \sum_{i=1}^n y_i^2)^{c+na} \theta_b^{2(c+na)-1} e^{-(d^2 + \sum_{i=1}^n y_i^2)\theta_b^2} \quad \theta_b > 0$$

. Dividing

$$\frac{p(\theta_a \mid y_{1:n})}{p(\theta_b \mid y_{1:n})}$$

and canceling out like terms (that don't depend on θ_a or θ_b) gives

$$\frac{p(\theta_a \mid y_{1:n})}{p(\theta_b \mid y_{1:n})} = \left(\frac{\theta_a}{\theta_b}\right)^{2(an+c)-1} e^{(\theta_b^2 - \theta_a^2)(d^2 + \sum_{i=1}^n y_i^2)}$$

. A sufficient statistic is $T(y_{1:n}) = \sum_{i=1}^n y_i^2$.

To find a sufficient statistic more formally using the likelihood, we use the Fisher-Neymar factorization theorem: if the probability density function is $p(y_{1:n} \mid \theta)$ where θ is a parameter, then T is sufficient for θ if and only if non-negative functions g and h can be found such that $p(y_{1:n} \mid \theta) = h(y_{1:n}) * g(\theta, T(y_{1:n}))$ (source= *Wikipedia*). Observe

$$\begin{aligned}
p(y_{1:n} | \theta) &= \prod_{i=1}^n \left[\frac{2}{\Gamma(a)} \theta^{2a} y_i^{2a-1} \exp(-\theta^2 y_i^2) \right] \\
&= \left[\left(\frac{2}{\Gamma(a)} \right)^n \prod_{i=1}^n y_i^{2a-1} \right] \left[\theta^{2an} \exp \left(-\theta^2 \sum_{i=1}^n y_i^2 \right) \right] \\
&= h(y_{1:n}) * g(\theta, T(y_{1:n})) = \sum_{i=1}^n y_i^2.
\end{aligned}$$

Therefore, a sufficient statistic is $T(y_{1:n}) = \sum_{i=1}^n y_i^2$.

I used ChatGPT to re-format the long equation above.

Part D

Determine $E[\theta|y_{1:n}]$.

In the introduction to Question 3, we are given

$$E[Y] = \frac{\Gamma(a + 1/2)}{\theta \Gamma(a)}$$

for the Galenshore distribution. Since the posterior is also Galenshore with updated parameters,

$$\theta | y_{1:n} \sim \text{Galenshore} \left(c + na, \sqrt{d^2 + \sum_{i=1}^n y_i^2} \right),$$

we have

$$E[\theta|y_{1:n}] = \frac{\Gamma((an + c) + 1/2)}{\sqrt{(d^2 + \sum_{i=1}^n y_i^2) \Gamma(an + c)}}.$$

Part E

Show that the form of the posterior predictive density is

$$p(y_{n+1}|y_{1:n}) = \frac{2y_{n+1}^{2a-1}\Gamma(an + a + c)}{\Gamma(a)\Gamma(an + c)} \frac{(d^2 + \sum y_i^2)^{an+c}}{(d^2 + \sum y_i^2 + y_{n+1}^2)^{(an+a+c)}}.$$

To calculate the posterior predictive density, we use

$$p(y_{n+1} \mid y_{1:n}) = \int_0^\infty p(y_{n+1} \mid \theta) p(\theta \mid y_{1:n}) d\theta.$$

The likelihood for the new observation is

$$p(y_{n+1} \mid \theta) = \frac{2}{\Gamma(a)} \theta^{2a} y_{n+1}^{2a-1} \exp(-\theta^2 y_{n+1}^2).$$

From Part A, the posterior density is

$$p(\theta \mid y_{1:n}) = \frac{2}{\Gamma(an+c)} \left(d^2 + \sum_{i=1}^n y_i^2 \right)^{an+c} \theta^{2(an+c)-1} \exp \left[-\theta^2 \left(d^2 + \sum_{i=1}^n y_i^2 \right) \right].$$

Substituting the likelihood and posterior into the posterior predictive integral gives

$$\begin{aligned} p(y_{n+1} \mid y_{1:n}) &= \int_0^\infty \left[\frac{2}{\Gamma(a)} \theta^{2a} y_{n+1}^{2a-1} \exp(-\theta^2 y_{n+1}^2) \right] \\ &\quad \times \left[\frac{2}{\Gamma(an+c)} \left(d^2 + \sum_{i=1}^n y_i^2 \right)^{an+c} \theta^{2(an+c)-1} \exp \left(-\theta^2 \left(d^2 + \sum_{i=1}^n y_i^2 \right) \right) \right] d\theta \\ &= \frac{4y_{n+1}^{2a-1}}{\Gamma(a)\Gamma(an+c)} \left(d^2 + \sum_{i=1}^n y_i^2 \right)^{an+c} \\ &\quad \times \int_0^\infty \theta^{2(an+a+c)-1} \exp \left[-\theta^2 \left(d^2 + \sum_{i=1}^n y_i^2 + y_{n+1}^2 \right) \right] d\theta. \end{aligned}$$

To obtain a proper posterior (integrates to 1 over the parameter space, Θ , we multiply and divide by

$$\frac{2}{\Gamma(an+a+c)} (d^2 + \sum_{i=1}^n y_i^2 + y_{n+1}^2)$$

, which gives

we obtain

$$p(y_{n+1} \mid y_{1:n}) = \frac{2y_{n+1}^{2a-1}\Gamma(an+a+c)}{\Gamma(a)\Gamma(an+c)} \frac{(d^2 + \sum_{i=1}^n y_i^2)^{an+c}}{(d^2 + \sum_{i=1}^n y_i^2 + y_{n+1}^2)^{an+a+c}}.$$

I used ChatGPT to re-format my long equations. Algebra is my own.